

COMPSYS704 20 LAB1

Introduction to Java programming

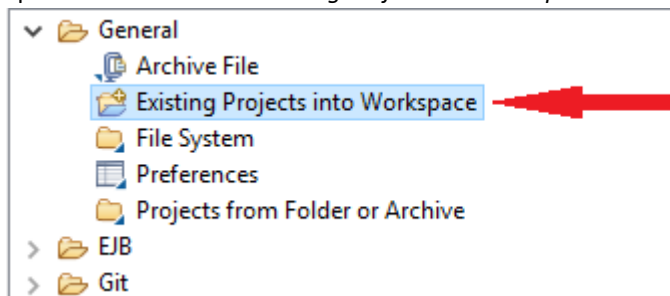
13/07/2023

Welcome to COMPSYS 704 Java lab. This lab introduces a set of basic Java programming exercises that will help you developing SystemJ programs that include Java statements and modules in Project 1. It is expected you are familiar with Java and able to write much more complex programs. In this lab we are using Eclipse Programming Environment (Eclipse IDE), which is also used for the development of SystemJ programs. However, SystemJ program compilation and execution can be done with command line interface, too.

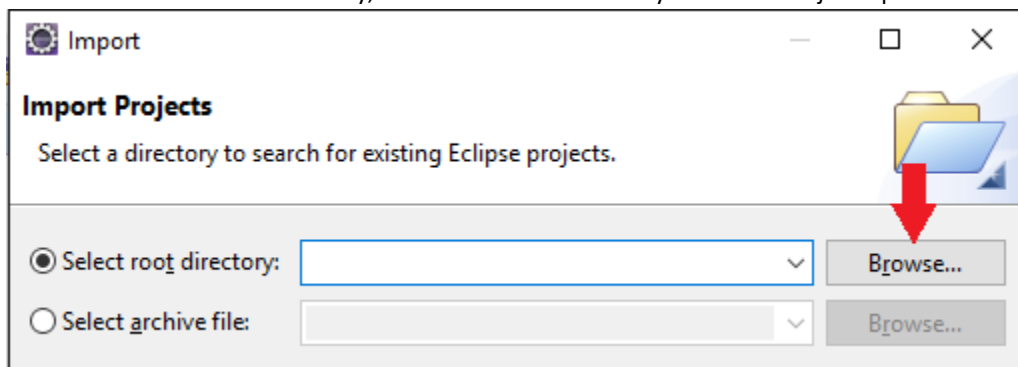
Setting up an Eclipse project

Most of the desktop machines in the undergraduate labs will have Eclipse IDE already installed. If not, download “Eclipse IDE for Java developers” at the official website <http://www.eclipse.org/downloads/eclipse-packages/>. We also already provided an Eclipse Java project for you; download **SystemJLab1Project.zip** from Canvas and extract the files to a directory of your choice. To import the project follow the steps below. You can apply this process for installing and using Eclipse IDE on your own computers (such as laptops, desktops).

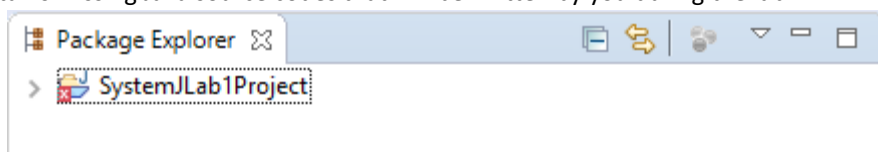
1. Run Eclipse (e.g. eclipse.exe).
2. From the menu bar, select *File* -> *Import* which will open a new Window where you can select different types of projects to import. Select *General* > *Existing Project into Workspace* as shown below. Click “Next”.



3. Select “Browse” and choose the directory, which is extracted from SystemJLab1Project.zip. Click “Finish”.

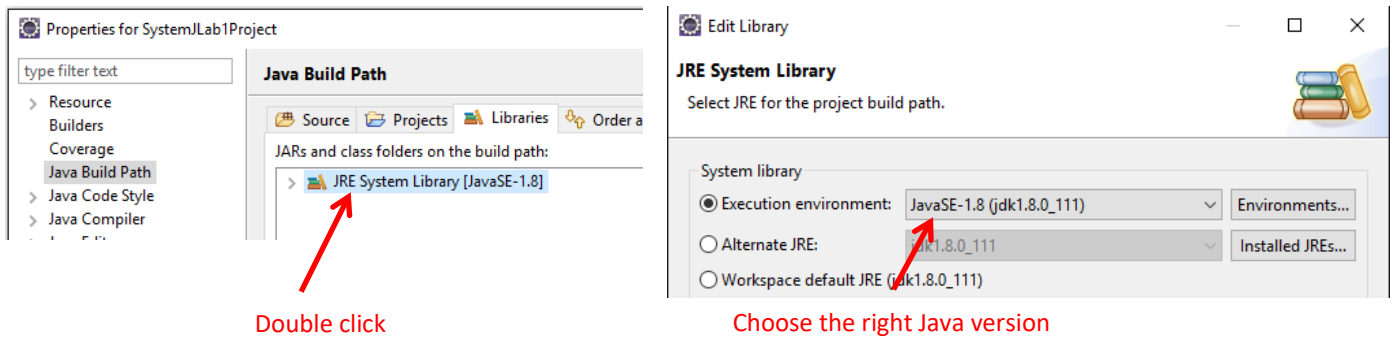


4. You will then see the Java project imported to your Project/Package explorer as shown below. Ignore the error icon for now, as it contains missing Java source codes that will be written by you during the lab.

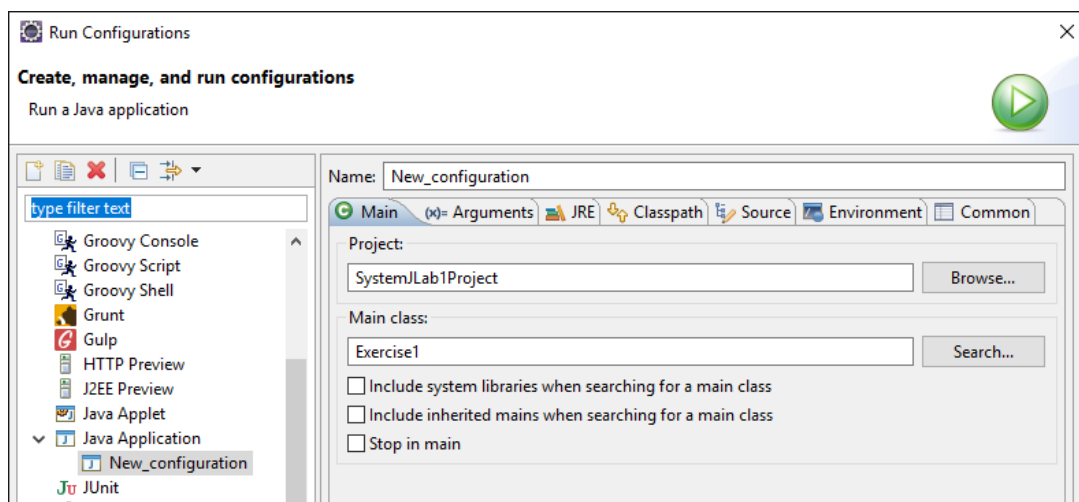


5. Make sure the imported project is using the right Java version. Right click the project folder and go to “**Properties**”. Click “**Java Build Path**” and check if the version of the “**JRE System Library**” matches with the JDK/JRE that is installed

on your machine. If the “JRE System Library” shows exclamation or “x” mark, double click the entry and update the Java library as shown below.



For your convenience, you are given template Java source codes for the Exercises 1-4. For the rest of the Exercises, you will need to create new Java classes as required by selecting *File -> New -> Class* on the menu bar or by right clicking “src” directory in the project explorer where you can add new Java classes. To run any Java program, select *File -> Run Configurations...* Double click “Java Application” and specify both Project and Main class that you want to run (see below). Click “Run” to run the Java program.



Exercise 1: In Exercise1.java, write the missing expressions in the arguments of each `println` method (in place of the “...”). Each expression should be written only using the variables `x` and `y`. When the program is run, it should print:

```
35
35!
The year is: 2015
The result is: 35
The result is: 300
```

```
public class Exercisel { // Each java file must have only one public class that has
    // the same name, in this case "Exercisel".
    public static void main(String[] args) {
        // there must be one main function for a public class, acting as the
        // entry point of the entire program.
        int x = 20;
        int y = 15;

        // play with the java standard output
        System.out.println(...);
        System.out.println(...+"!");
        System.out.println("The year is: " + ...);
        System.out.println("The result is: " + ...);
        System.out.println("The result is: " + ...);
    }
}
```

Exercise 2: Implement the `Fibonacci` class in `Exercise2.java` so that the program prints correct Fibonacci sequences for 10 and 20. Fibonacci number is sum of previous two Fibonacci numbers $fn = fn-1 + fn-2$. First 10 Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, 34, 55.

```
public class Exercise2 {
    public static void main(String[] args) {
        // the following two listx objects uses different constructors.
        Fibonacci list1 = new Fibonacci(10);
        Fibonacci list2 = new Fibonacci(20);
        list1.printList();
        list2.printList();
    }
    class Fibonacci {
        // Fibonacci class
    }
}
```

Exercise 3: In `Exercise3.java`, implement the classes `Shape`, `Circle`, and `Square` according to the UML diagram shown in Figure 1. Implement methods so the given code makes sense.

```
import shapes.*;
public class Exercise3 {
    public static void main(String[] args) {
        Circle circle1 = new Circle("Circle 1", 3);
        Square square1 = new Square("Square 1", 3);
        System.out.println(circle1.getName()+"'s area is: "+circle1.calculateArea());
        System.out.println(square1.getName()+"'s area is: "+square1.calculateArea());
    }
}
```

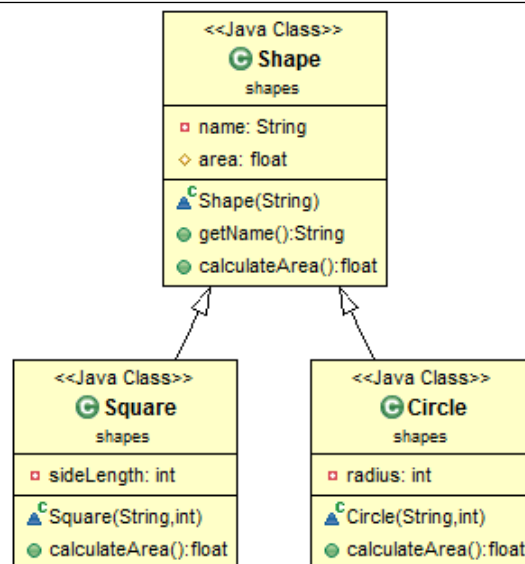


Figure 1. UML diagram for Exercise 3

Exercise 4: In Exercise4.java, specify the parameter of the third `call` method so that when the program is executed it prints `call3`.

```
package shapes;
import java.util.List;
public class Exercise4 {
    public static void call(Object x){
        System.out.println("call1");
    }
    public static void call(List x){
        System.out.println("call2");
    }
    public static void call(...){
        System.out.println("call3");
    }

    public static void main(String[] args) {
        call(null);
    }
}
```

Exercise 5:

- (a) Write a **method** that takes an integer value and returns the number with its digits reversed. (For example, if number 1938 is given, the **method** should return 8391).
- (b) Write a Java application which uses the method from (a). This application should read a value from the keyboard and display the result. (It should also print a proper error message if an exception occurs when a wrong type input is given from the keyboard).

Exercise 6: Dates are printed in several common formats. Two of the more common formats are (a) 05/23/1945 and (b) 23 May 1945. Write a Java application that reads a date from keyboard in format (a) and prints it in format (b).

Exercise 7:

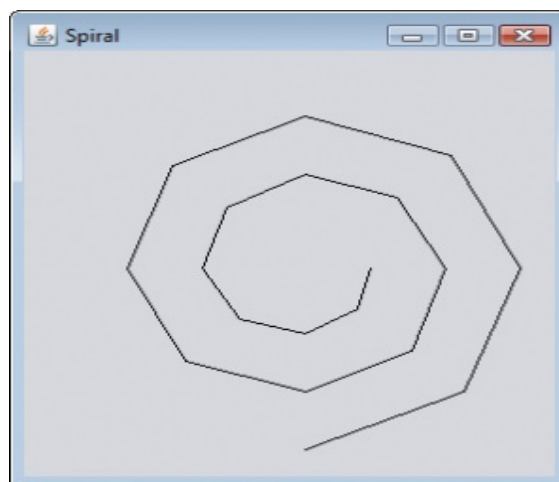
- (a) Write the Java code for a class called `MyExceptionGen`. Your code should get characters from the keyboard and generate an exception when any numeric keyboard character (i.e. 0 to 9) is entered, terminate the program with the exception message as **"NoNumbersAllowed"**.
- (b) Re-write the code in a new way which can catch that exception and continue reading from the keyboard until **"Esc"** key is pressed. The characters read should be printed on screen (but converting the number characters to `*`). For example, if in this case the user enters `"Akxd9zD2MP"` the printed result would be `"Akxd*zD*MP"`.

Exercise 8: Write a Java program that reads a sentence from the keyboard and prints each word in the sentence and the number of vowels in that word. (Vowels are a, e, i, o, u). (Each word followed by the number of its vowels should be printed in a separate line).

Exercise 9: Implement the following pseudo-code Bubble sorting algorithm in Java.

```
procedure bubbleSort( A : list of sortable items )
    n = length(A)
    repeat
        swapped = false
        for i = 1 to n-1 inclusive do
            /* if this pair is out of order */
            if A[i-1] > A[i] then
                /* swap them and remember something changed */
                swap( A[i-1], A[i] )
                swapped = true
            end if
        end for
    until not swapped
end procedure
```

Exercise 10: Write a Java program to draw a spiral similar to the one shown in the following figure.



Here we list some of good tutorial websites on various topics of Java including basic Java concepts and Swing GUI library. For those who need additional materials to get familiarized with Java, these websites are good starting point:

Java tutorial - useful for everyone

<http://www.tutorialspoint.com/java/>

Java Swing tutorial

<http://www.tutorialspoint.com/swing/>

also, you can find Oracle's official websites:

Java

<https://docs.oracle.com/javase/tutorial/index.html>

Java Swing

<https://docs.oracle.com/javase/tutorial/uiswing/index.html>